

```
use vivek;
```

```
select emp.emp_id, emp.emp_name, dept.dept_name , dept.location from
employees as emp inner join
departments as dept
on emp.dept_id = dept.dept_id; -- 1
```

emp_id	emp_name	dept_name	location
101	Amit	HR	Jaipur
102	Bhavna	IT	Bangalore
103	Chirag	IT	Bangalore
104	Divya	Finance	Mumbai
106	Farhan	Marketing	Delhi
108	Hina	Finance	Mumbai

```
select proj.proj_id, proj.proj_name,dept.dept_name,proj.budget from
projects as proj inner join
departments as dept
on proj.dept_id = dept.dept_id; -- 2
```

proj_id	proj_name	dept_name	budget
1	Payroll Revamp	HR	120000
2	Cloud Migration	IT	500000
3	Mobile App	IT	300000
4	Audit Automation	Finance	250000
5	Brand Refresh	Marketing	180000

```
select emp.emp_name,dept.dept_name,proj.proj_name from employees as emp
inner join
projects as proj on emp.dept_id = proj.dept_id inner join departments
as dept
on dept.dept_id = proj.dept_id ; -- 3
```

emp_name	dept_name	proj_name
Amit	HR	Payroll Revamp
Bhavna	IT	Mobile App
Bhavna	IT	Cloud Migration
Chirag	IT	Mobile App
Chirag	IT	Cloud Migration
Divya	Finance	Audit Automation
Farhan	Marketing	Brand Refresh
Hina	Finance	Audit Automation

```
select emp.emp_id,emp.emp_name,emp.dept_id,dept.dept_name from
employees as emp left join
departments as dept
on emp.dept_id = dept.dept_id; -- 4
```

emp_id	emp_name	dept_id	dept_name
101	Amit	10	HR
102	Bhavna	20	IT
103	Chirag	20	IT

emp_id	emp_name	dept_id	dept_name
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104	Divya	30	Finance
105	Esha		
106	Farhan	40	Marketing
107	Gaurav	60	
108	Hina	30	Finance

```
select dept.dept_id,dept.dept_name,proj.proj_name,proj.budget from
departments as dept left join
projects as proj
on dept.dept_id = proj.dept_id; -- 5
```

dept_id	dept_name	proj_name	budget
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10	HR	Payroll Revamp	120000
20	IT	Mobile App	300000
20	IT	Cloud Migration	500000
30	Finance	Audit Automation	250000
40	Marketing	Brand Refresh	180000
50	Legal		

```
select emp.emp_id,emp.emp_name,emp.dept_id from employees as emp left
join
departments as dept
on emp.dept_id = dept.dept_id where dept.dept_name is null; -- 6
```

emp_id	emp_name	dept_id
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105	Esha	
107	Gaurav	60

```
select emp.emp_name,dept.dept_id,dept.dept_name from employees as emp
right join departments as dept
on emp.dept_id = dept.dept_id; -- 7
```

emp_name	dept_id	dept_name
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Amit	10	HR
Chirag	20	IT
Bhavna	20	IT
Hina	30	Finance
Divya	30	Finance
Farhan	40	Marketing
	50	Legal

```
select d.dept_name , p.proj_id, p.proj_name, p.dept_id from departments
as d
right join projects as p
on d.dept_id = p.dept_id; -- 8
```

dept_name	proj_id	proj_name	dept_id
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HR	1	Payroll Revamp	10
IT	2	Cloud Migration	20
IT	3	Mobile App	20

	dept_name	proj_id	proj_name	dept_id
Finance	4		Audit Automation	30
Marketing	5		Brand Refresh	40
	6		Data Lake	70

```
-- Q9. Find departments that currently have NO employee assigned.
select d.dept_id, d.dept_name , d.location from departments as d
left join employees as e
on d.dept_id = e.dept_id where e.emp_name is null;
```

	dept_name	proj_id	proj_name	dept_id
HR	1		Payroll Revamp	10
IT	2		Cloud Migration	20
IT	3		Mobile App	20
Finance	4		Audit Automation	30
Marketing	5		Brand Refresh	40
	6		Data Lake	70

```
-- Q10. Produce every possible pairing of the Finance/Legal departments
with employees
-- earning above 70000.
```

```
select e.emp_name ,e.salary , d.dept_name from employees as e
cross join departments as d
where (dept_name = 'legal' or dept_name = 'Finance') and salary >70000;
```

	emp_name	salary	dept_name
Bhavna	72000	Legal	
Bhavna	72000	Finance	
Hina	83000	Legal	
Hina	83000	Finance	

```
-- Q11. Pair every project having a budget of at least 400000 with
every department
```

```
-- located in Mumbai or Pune.
```

```
select p.proj_name,p.budget,d.dept_name,d.location from projects as p
cross join departments as d
where budget >= 400000 and d.location in("mumbai","pune");
```

	proj_name	budget	dept_name	location
Data Lake	400000	Finance	Mumbai	
Cloud Migration	500000	Finance	Mumbai	
Data Lake	400000	Legal	Pune	
Cloud Migration	500000	Legal	Pune	

```
-- Q12. How many total row combinations result from CROSS JOIN
-- of employees and departments? Show the count
select count(*) as total_combiantion from employees cross join
departments;
```

	total_combiantion
	40

```
-- Q13. Show employees whose salary is greater than 60000.
select e.emp_id,e.emp_name,d.dept_id , e.salary, e.city from employees
as e
left join departments as d
on e.dept_id = d.dept_id where salary > 60000;
```

emp_id	emp_name	dept_id	salary	city
102	Bhavna	20	72000	Bangalore
103	Chirag	20	68000	Bangalore
106	Farhan	40	61000	Delhi
108	Hina	30	83000	Mumbai

```
-- Q14. Show all employees based in Delhi or Mumbai.
select e.emp_id,e.emp_name,e.salary,e.city from employees as e left
join departments as d
on e.dept_id = d.dept_id where city in ('delhi','mumbai');
```

emp_id	emp_name	salary	city
104	Divya	55000	Mumbai
105	Esha	39000	Delhi
106	Farhan	61000	Delhi
108	Hina	83000	Mumbai

```
-- Q15. Show employees whose dept_id is NULL (unassigned employees).
select e.emp_id , e.emp_name, e.emp_id , e.salary from employees as e
left join
departments as d
on e.dept_id = d.dept_id where e.dept_id is null;
```

emp_id	emp_name	emp_id	salary
105	Esha	105	39000

```
-- Q16. Show departments located in Bangalore or Delhi.
select distinct d.dept_id,d.dept_name,d.location from departments as d
left join
employees as e on d.dept_id = e.dept_id
where location in('bangalore','delhi');
```

dept_id	dept_name	location
20	IT	Bangalore
40	Marketing	Delhi

```
-- Q17. Show departments whose dept_id is 30 or higher.
select distinct d.dept_id,d.dept_name,d.location from departments as d
left join employees as e on d.dept_id = e.dept_id
where d.dept_id >=30;
```

dept_id	dept_name	location
30	Finance	Mumbai
40	Marketing	Delhi
50	Legal	Pune

```
-- Q18. Show departments whose name starts with the letter "F" or "L".
select distinct d.dept_id, d.dept_name, d.location from employees as e
right join departments as d on e.dept_id = d.dept_id
where d.dept_name like 'f%' or d.dept_name like 'l%';
```

dept_id	dept_name	location
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30	Finance	Mumbai
40	Marketing	Delhi
50	Legal	Pune

```
-- Q19. Show projects with a budget between 200000 and 400000
(inclusive)
select distinct p.proj_id , p.proj_name,p.dept_id, p.budget from
projects as p
left join departments as d on p.dept_id = d.dept_id
where p.budget between 200000 and 400000;
```

proj_id	proj_name	dept_id	budget
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3	Mobile App	20	300000
4	Audit Automation	30	250000
6	Data Lake	70	400000

```
-- Q20. Show all projects that belong to dept_id 20.
select distinct p.proj_id , p.proj_name,p.dept_id, p.budget from
projects as p left join
employees as e on p.dept_id = e.dept_id where p.dept_id = '20';
```

proj_id	proj_name	dept_id	budget
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2	Cloud Migration	20	500000
3	Mobile App	20	300000

```
-- Q21. Show projects whose name contains the word "a" and budget is
under 300000.
```

```
select distinct p.proj_id , p.proj_name, p.budget from projects as p
left join
employees as e on p.dept_id = e.dept_id
where proj_name like '%a%' and budget < 300000;
```

proj_id	proj_name	budget
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1	Payroll Revamp	120000
4	Audit Automation	250000
5	Brand Refresh	180000

```
-- Q22. Show employee name, department name, location and project name
for all matching rows
```

```
-- across all three tables.
```

```
select e.emp_name, d.dept_name, d.location ,p.proj_name, p.budget from
employees as e
inner join departments as d inner join projects as p
on d.dept_id = e.dept_id and p.dept_id = e.dept_id;
```

emp_name	dept_name	location	proj_name	budget
Amit	HR	Jaipur	Payroll Revamp	120000
Bhavna	IT	Bangalore	Mobile App	300000
Bhavna	IT	Bangalore	Cloud Migration	500000
Chirag	IT	Bangalore	Mobile App	300000
Chirag	IT	Bangalore	Cloud Migration	500000
Divya	Finance	Mumbai	Audit Automation	250000
Farhan	Marketing	Delhi	Brand Refresh	180000
Hina	Finance	Mumbai	Audit Automation	250000

-- Q23. Show ALL employees, plus department and project details where available

-- (keep employees even with no dept/project).

```
select e.emp_id , e.emp_name, d.dept_name , p.proj_name from employees
as e left join
departments as d on e.dept_id = d.dept_id left join projects as p
on d.dept_id = p.dept_id;
```

emp_id	emp_name	dept_name	proj_name
101	Amit	HR	Payroll Revamp
102	Bhavna	IT	Mobile App
102	Bhavna	IT	Cloud Migration
103	Chirag	IT	Mobile App
103	Chirag	IT	Cloud Migration
104	Divya	Finance	Audit Automation
105	Esha		
106	Farhan	Marketing	Brand Refresh
107	Gaurav		
108	Hina	Finance	Audit Automation

-- Q24. Show employees earning more than 60000 along with department and any project

-- over 250000 budget.

```
select e.emp_name,e.salary, d.dept_name, p.proj_name , p.budget from
employees as e
left join departments as d on e.dept_id = d.dept_id
left join projects as p on d.dept_id = p.dept_id where salary >60000
and budget >250000;
```

emp_name	salary	dept_name	proj_name	budget
Bhavna	72000	IT	Mobile App	300000
Bhavna	72000	IT	Cloud Migration	500000
Chirag	68000	IT	Mobile App	300000
Chirag	68000	IT	Cloud Migration	500000

-- Q25. List ALL departments with their employees and projects,
-- including departments having neither.

```
select d.dept_id, d.dept_name , e.emp_name , p.proj_name from
departments as d left join
employees as e on d.dept_id = e.dept_id left join
```

```
projects as p on d.dept_id = p.dept_id;
```

	dept_id	dept_name	emp_name	proj_name
10	HR		Amit	Payroll Revamp
20	IT		Chirag	Mobile App
20	IT		Chirag	Cloud Migration
20	IT		Bhavna	Mobile App
20	IT		Bhavna	Cloud Migration
30	Finance		Hina	Audit Automation
30	Finance		Divya	Audit Automation
40	Marketing		Farhan	Brand Refresh
50	Legal			

```
-- Q26. Show employees who work in a department located in Bangalore or  
Mumbai, along with the projects of that  
-- department.
```

```
select e.emp_name, d.location, p.proj_name from employees as e left  
join  
departments as d on d.dept_id = e.dept_id left join  
projects as p on d.dept_id = p.dept_id where d.location in(  
'bangalore','mumbai');
```

	emp_name	location	proj_name
Bhavna	Bangalore	Mobile App	
Bhavna	Bangalore	Cloud Migration	
Chirag	Bangalore	Mobile App	
Chirag	Bangalore	Cloud Migration	
Divya	Mumbai	Audit Automation	
Hina	Mumbai	Audit Automation	

```
-- Q27. Pair employees who work in the same city (avoid duplicate pairs  
and self-pairing).
```

```
select e1.emp_name as employees_1 , e2.emp_name as employees_2, e1.city  
from employees as e1 join employees as e2  
on e1.city = e2.city and e1.emp_id < e2.emp_id;
```

	employees_1	employees_2	city
Bhavna	Chirag	Bangalore	
Divya	Hina	Mumbai	
Esha	Farhan	Delhi	

```
-- Q28. Show every project along with employees of that department;  
keep projects with no employees.
```

```
select p.proj_name , p.dept_id , e.emp_name from projects as p left  
join  
employees as e on p.dept_id = e.dept_id;
```

proj_name	dept_id	emp_name
Payroll Revamp	10	Amit

proj_name	dept_id	emp_name
Cloud Migration	20	Chirag
Cloud Migration	20	Bhavna
Mobile App	20	Chirag
Mobile App	20	Bhavna
Audit Automation	30	Hina
Audit Automation	30	Divya
Brand Refresh	40	Farhan
Data Lake	70	